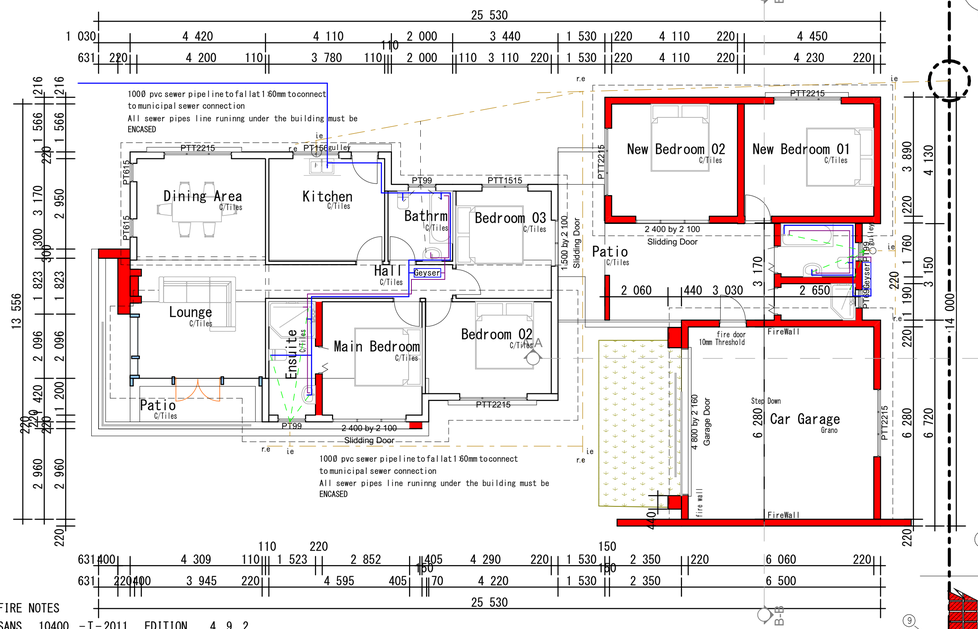
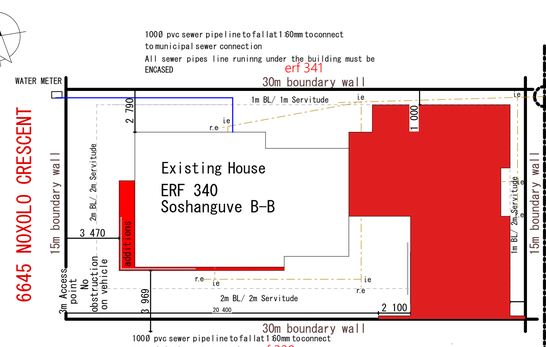
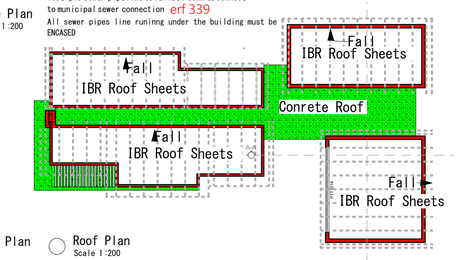


Ground Floor Plan
Scale 1:100



Site Plan
Scale 1:200



Roof Plan
Scale 1:200

WALL LEGEND
scale 1:100

- New Wall
- Existing Wall
- Demolishing Wall

SANS 10400 XA
WALLS :
Non - masonry wall will have R values as provided SANS 10400 - XA - 4.4.3.1)
Double sking masonry with plaster inside or render outside complies Single leaven 140mm with plaster inside and render outside complies SANS 10400 - XA - 4.4.3.2)
Other masonry walls will have R value of 0.5. SANS 10400 - XA - 4.4.3.3)

HEALTH :

Ceiling level to be at least 2.4m in habitable rooms or comply with 6 of the NBR
Window to be at least 10% and opening section 5% of the floor area of the habitable area
Water hating and cooking apparatus in house and out building to be smoke free

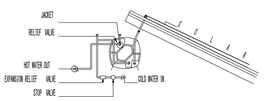
ROOF NOTES

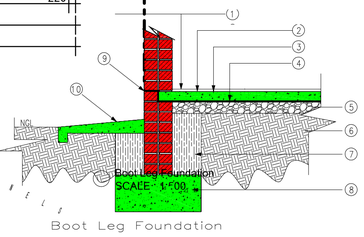
IBR Sheeting Laid to Specification
IBR Sheets at 8 degrees on 38mm x 38mm Battens SABS aproval Grade S5 @ 320mm c /c with plastic underlay
152mm x 38mm Rafters SABS aproved Grade S5 @ 760mm c /c tied with 114mm x 38mm Web members SABS aproved Grade S 5 @ 760mm c /c.
Tied beam 152mm x 38mm SABS aproved Grade S 5 and 76mm Wall plate Grade S5 and ceiling to be nailed on 38mm Brendering @ 450mm c

FIRE NOTES
SANS 10400 -T-2011 EDITION 4.9.2
any door between such garage & any such room shall have a fire resistance of not that 30 mm & such doorway shall require threshold of not les than 10 mm
no combustable roof shall penetrate the separating element diving the space between the garage & habitable room

ROOF NOTES

IBR Sheeting Laid to Specification
IBR Sheets at 8 degrees on 38mm x 38mm Battens SABS aproval Grade S5 @ 320mm c /c with plastic underlay
152mm x 38mm Rafters SABS aproved Grade S5 @ 760mm c /c tied with 114mm x 38mm Web members SABS aproved Grade S 5 @ 760mm c /c.
Tied beam 152mm x 38mm SABS aproved Grade S 5 and 76mm Wall plate Grade S5 and ceiling to be nailed on 38mm Brendering @ 450mm c

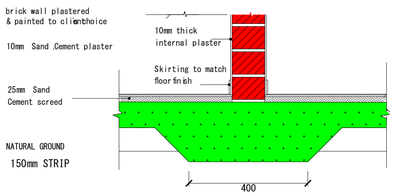




Boot Leg Foundation

BOOT LEG FOUNDATION DETAILS

01 - 76mm Meranti skirting with 19mm quadrant.
02 - Finished floor level
03 - 125mm Conrete surface bed
04 - 250mm Micro black DPC
05 - 3x150mm Hardcore filling (G7) compacted to 93% mod AASHTO density
06 - Compact in sub material 93% mod AASHTO density for cut & fill compact 150mm layers to 93% mod AASHTO density
07 - 60mm x 250mm Thick strip concrete footing
09 - 250mm Micro black DPC
10 - 1m Apron with Refl 193 and expension joint for every 2m or less



110 mm STRIP FOUNDATION DETAIL
Scale 1:100

FENESTRATION :
FENESTRATION CALCULATIONS AND NATURAL ENVIRONMENTAL CONTROL (SANS 10400 XA)

Total of Floor Area 147.96m²
Total of Glazing Area 22.83m²
Total 15% of floor Area 21.05m²

Therefore glazing area is more than 15% of the area of proposal therefore calculations i.t.o SANS 204 is required.

HOT WATER CALCULATIONS :

Volume of annual hot water heating requirements
Population calculations

Bed rooms	People per bed rm	Total
3	2	9

Volume of annual hot water heating requirements
Population calculations

DEMAND :	115OL/UNIT	HEATER POWER
80L /Person/day	115OL_960= 1.1979	3kW units: 0X9 724W /day

Thus : 3840kw /Year but use of solar panel 60% saving
Thus : 1920kw /Year hot water heating requirement

ROOF ASSEMBLIES (R -VALUE):

R -value= 3.7m²K /W	Climantic zone 4
Direction of Heat flow - UP	
Basic roof assemblies	Clay roman tiles
R_Value for roof covering	0.35m²K /W
R_Value for ceiling ins	0.05m²K /W
Required added R-Value inst	3.3Km²/W

ROOF UNVENTIALATED

Out door air m (7m/s)	0.03
Roof clay tiled 922kg/m³)	0.02
Roof air space non -reflective	0.18
Plasterboard gypsum (10mm 880kg/m³)	0.06
Indoor air still air	0.11
Generic Insulation Product added fibreglass blanket	
Required added R value inst	0.4
Total R value (3.3 + 0.4)	3.1

ENERGY CONSUMPTION LIGHTING
BUILDING CLASSIFICATIONS : H4

Legend / Types (LOS 005)	Amount	Calculations	Total watts
Ceiling Lights	06	06 x 11w	66 watts
Ceiling Lights	02	02 x 22w	44watts
Down Lights	80	80x 6w	480watts
Fluorescent Lights (KB 22GE)	02	02 x 18w	36 watts
Total Lights			626 watts

INTERNAL ACTUAL :
Permissible 5w /m² or 5Kwh .a (a= year) assume lights are on from 07.00 to 22.00 each day
5h/day= 52 (weeks) x7 (days) = 1820h.a

ENERGY CONSUPTION :

Allowed 5kWh /sqm per year	Total Residence	626
5Wh x 329.52sqm = 1,647.6m².a	House area	329

TOTAL INTERNAL AND EXTERNAL ACTUAL :
1.139.32kw < 1,647.6Kwh .a (per year)

Consumption : The design complies with the requirement of SANS 0400_ XA 2011 for energy consumption

Max Demand as per SANS 10142 = 5W per sqm.
Actual : 626w / 329.52sqm = 1.9w/m²

GENERAL NOTES :
All work to be in accordance with the national building regulations SANS 10400-2011
All levels and dimensions to be checked and verified on site prior to commencement of work any discrepancies to be reported to the architect immediatly

FOUNDATION NOTES :
All foundations foundation walls retaining walls, sub soil stormwater drainage and soil compaction to 'Engs spec& approval
230mm walls on min 600 (w) x300 (h) & 115mm walls on min 450 (w) x300 (h) strip foundations with 20mpa concrete

SURFACE BED NOTES :
Internal surface bed 100mm 20mpa concrete on dpc& well compacted lat 150mm layers.
Slabs are not to be cast on in situ material allow to remove min 300mm and re compact in 150mm layers external surface bed 85mm 20mpa concrete as above

WALL NOTES :
Brickwork Clay stock keyed to receive plasterinsh
230mm brick walls with brickforce every 4 courses and every 6 course for courses over openings or to 'Engs spec & approval
All brickwork min 4mpa with Class II mortar

ROOF NOTES :
Refer to roof notes

GLAZING NOTES :
All glazing to comply with "part N OF SANS 10400- 2011."

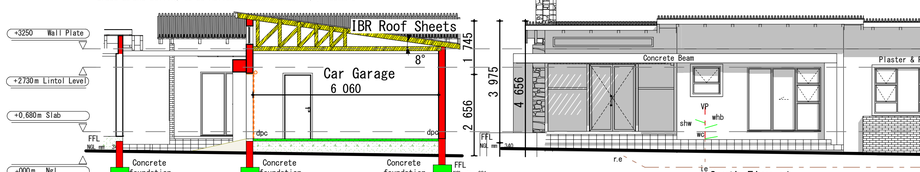
PLUMBING & DRAINAGE :
All plumbing to comply with P of the SANS 10400 :2011
All waste pipes and drains to be accessible along their entire length
Provide inspection eyes at all drain bends and juctions and at a mix
Cleaning eyes to be provided at all bends and juctions of waste pipes
Waste pipes to be tted with:
mm resealtraps
Water connections to be tted with
mm dia to WHB ,WC and Showers
mm dia to Geyser
mm dia to FHR
All vent pipes to discharge to external air
mm dia waste pipes require to 75mm dia sleeve
110mm dia waste pipes require a 150mm dia sleeve
For plumbing and drainage layout plans and cross sections refer to registed wet services
engineers drawings for rational design drawings . 10400
All sorm water to be collected and drained to comply with part P of the SANS drain municipal storm water drainage system

STRUCTURAL :
All reinforced concrete work and retaining walls to be strictly in accordance with registered structural engineers details and specifications

MECHANICAL VENTILATION & LIGHTING :
All lighting and ventilation (fresh air) and extrct to comply with part O SANS 10400 accordance with registered mechanical engineers drawings and registerd electrical engineer drawings
All articial lighting to comply with SANS 10400 XA

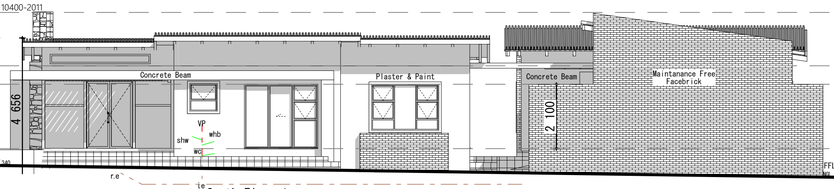
SANS 10400 XA (WALLS :)
Non masonary walls will have R" values as provided (SANS 10400 XA 4.4.3.1)
Double sking masonary with plaster inside or render outside to complies of Single leafe min
150mm with plaster inside or render outside to complies SANS 10400 XA 4.4.3.2)
Other masonary walls will have R' value of 0.35 SANS 10400 XA 4.4.3.3)

REV :	DESCRIPTION :	DATE :



Section A- A
Scale 1:100

Floor covering as per schedule. 30mm cement screed on 85mm concrete surface
mm thick layer of G5 material or better compacted to 95% mod AASHTO density 250mm micron polyonlefin sheeting under floor slab on well compacted filling in layers not exceeding 150mm treated with SABS approved poisoned weed acc to SABS0124.

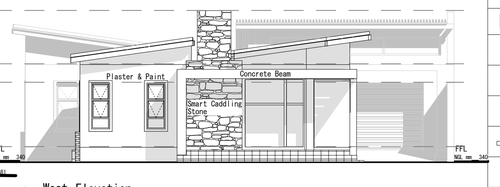


South Elevation
Scale 1:100

1000 pvc sewer pipeline to fall at 1:60mm to connect to municipal sewer connection
All sewer pipes line runinng under the building must be ENCASED

ROOF NOTES

IBR Sheeting Laid to Specification
IBR Sheets at 8 degrees on 38mm x 38mm Battens SABS aproval Grade S5 @ 320mm c /c with plastic underlay
152mm x 38mm Rafters SABS aproved Grade S5 @ 760mm c /c tied with 114mm x 38mm Web members SABS aproved Grade S 5 @ 760mm c /c.
Tied beam 152mm x 38mm SABS aproved Grade S 5 and 76mm Wall plate Grade S5 and ceiling to be nailed on 38mm Brendering @ 450mm c



West Elevation
Scale 1:100

Electrical Legend

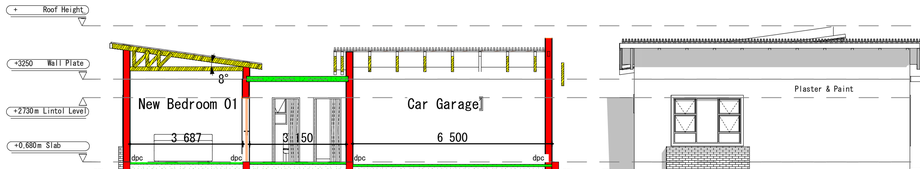
- Distribution board
- Down lights
- Ceiling lights
- Double wall plugs
- Flourescent Lights
- Double lights switch
- Singlelights switch
- Stove point
- 300L Geyser in roof spec

PROJECT TITLE
PROPOSED ADDITIONS
ERF 340 Soshanguve B-B
Pretoria
0152
For Matlakala Jane Moela

Owner /Client Sig :
Architect Sig :
Engineer Sig :

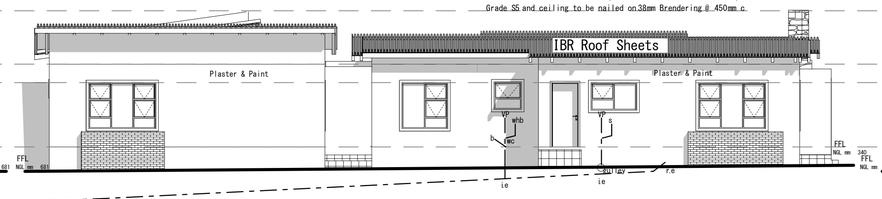
AREA SCHEDULE

SITE AREA (erf)	:	450 m²
EXISTING GROUND FLOOR AREA	:	115 m²
ADDITIONAL FLOOR AREA	:	138 m²
	:	
GRANT TOTAL FLOOR AREA	:	245 m²
PERMISSIBLE COVERAGE	:	60%
ACTUAL COVERAGE	:	56%



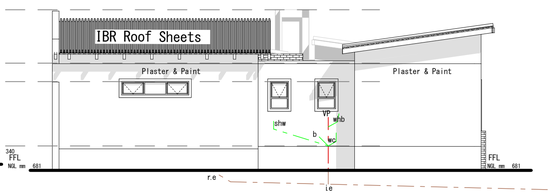
Section B- B
Scale 1:100

Floor covering as per schedule. 30mm cement screed on 85mm concrete surface
mm thick layer of G5 material or better compacted to 95% mod AASHTO density 250mm micron polyonlefin sheeting under floor slab on well compacted filling in layers not exceeding 150mm treated with SABS approved poisoned weed acc to SABS0124.



North Elevation
Scale 1:100

1000 pvc sewer pipeline to fall at 1:60mm to connect to municipal sewer connection
All sewer pipes line runinng under the building must be ENCASED



East Elevation
Scale 1:100

1000 pvc sewer pipeline to fall at 1:60mm to connect to municipal sewer connection
All sewer pipes line runinng under the building must be ENCASED

REVISIONS

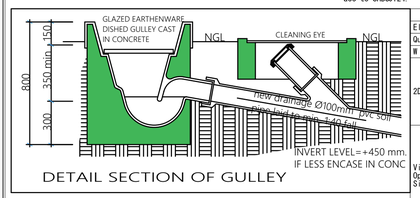
REV	DATE	DETAILS	BY
A			

MAKENA BUILDING GROUP
Contact: General Manager
Cell: (069) 533 7778
4/1619 Soshanguve A
Soshanguve ,0152

Designed by: Kgothatso Makena
Checked by :
SACAP :
signature

ISSUED FOR OWNER : ☑ ISSUED FOR CONSTRUCTION : ☐ ISSUED FOR TENDER : ☐

Job No. 2024-05
Drawing No. -A-01
Rev. No. ###.##.##
Sheet Size A1
Issue Date



DETAIL SECTION OF GULLEY

Window Schedule

Element ID	CWD - 022	CWD - 022	PT69	PT99	PT99	PT156	PT615	PTT1515	PTT1518	PTT1518	PTT2215	PTT2215
Quantity	1	1	1	1	2	1	2	1	1	1	1	4
W x H Size	1 448 × 1 500	900 × 1 500	600 × 900	600 × 900	900 × 900	1 500 × 900	600 × 1 500	1 500 × 1 500	3 273 × 312	5 559 × 600	2 200 × 450	2 200 × 1 500
2D Symbol												
View from Side Opposite to Opening Side												

Door Schedule

Element ID	DOO - 023	Garage Door	New Miranthi Door	Slidding Door	Slidding Door
Quantity	1	1	8	1	2
W x H Size	2 800×2 100	4 800×2 160	880×2 100	1 500×2 100	2 400×2 100
2D Symbol					
View from Side Opposite to Opening Side					